B. Sc. in CSE, UAP

items / objects :	1	2	3	4
weight :	3	2	5	4
value / profits :	4	3	6	5

max weight = 5

n = 4

1 kg

1.5 kg

3 kg

0

B (i ↓, w →)	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0					

But since the past value/ profit is greater we take the max

Here, it was supposed to be 6

items / objects :	1	2	3	4
weight :	3	2	5	4
value / profits :	4	3	6	5

max weight = 5

n = 4

1kg

1.5 kg

3kg

(3+4)

5-2= 3

B

i ↓ \ w →	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0				
4	0					

items / objects : 1 2 3 4
weight : 3 2 5 4
value / profits : 4 3 6 5

max weight = 5
$n = 4$

3kg

$i \backslash w$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

Max value we can take

Weight : 3 2

Value/ Profits : 4 3 6 5

B \ i ↓ → w	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

$$B[i,w] = \max(B[i-1,w],\ B[i-1,\ w-w[i]] + v[i])$$

$$B[4,4] = \max(B[3,4],\ B[3,4-4]+5)$$

$$= \max(4,\ B[3,0]+5)$$

$$= \max(4,\ 0+5=5)$$

$$= \max(4,5)$$

Value/ Profits: 4 3 6 5

B \ W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

item: 1 2 3 4

Backtracking

if we take the item, we set it 1
otherwise we set is 0

$$B[i,w] = max(B[i-1,w], B[i-1, w-w[i]] + v[i])$$

$$B[4,4] = max(B[3,4], B[3, 4-4] + 5)$$

$$= max(4, \ [3,0] + 5)$$

$$= max(4,$$

$$= max(4,$$

$$B[4,5] = max(B[3,5], B[3, 5-4] + 5)$$

$$= max(B[3,5), B[3,1] + 5)$$

$$= max(7, B\ 0 + 5 = 5)$$

$$= max(7, 5)$$

$$= 7$$

Weight : 3 2 5 4

Value/ Profits : 4 3 6 5

n = 4

1kg

15 kg

3kg

item: 1 2 3 4

0

B \ W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
x 4	0	0	3	4	5	$\boxed{7}$

Since the previous value is equal to current value we do not take the corresponding item of current value

$$B[i,w] = \max(B[i-1,w],\ B[i-1,\ w-w[i]] + v[i])$$

$$B[4,4] = \max(B[3,4],\ B[3,4-4]+5)$$
$$= \max(4,\ B[3,0]+5)$$
$$= \max(4,\ 0+5=5)$$
$$= \max(4,5)$$
$$= 5$$

$$B[4,5] = \max(B[3,5],\ B[3,5-4]+5)$$
$$= \max(B[3,5),\ B[3,1]+5)$$
$$= \max(7,\ B\ 0+5=5)$$
$$= \max(7,5)$$

Weight: 3 2 5 4

$n = 4$

Value/ Profits: 4 3 6 5

1kg 15 kg 3kg

item: 1 2 3 4

0 0

B	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
x 3	0	0	3	4	4	7
x 4	0	0	3	4	5	$\boxed{7}$

$$B[i,w] = \max(B[i-1,w],\ B[i-1,\ w-w[i]] + v[i])$$

$$B[4,4] = \max(B[3,4],\ B[3,4-4]+5)$$
$$= \max(4,\ B[3,0]+5)$$
$$= \max(4,\ 0+5=5)$$
$$= \max(4,5)$$
$$= 5$$

$$B[4,5] = \max(B[3,5],\ B[3,5-4$$
$$= \max(B[3,5],\ B[3,1]$$
$$= \max(7,\ B\ 0+5$$
$$= \max(7,5)$$

Weight : 3 2 5 4

Value / Profits : 4 3 6 5

$n = 4$

1kg

15 kg

3kg

$i \downarrow$ \ B $\overset{w}{\rightarrow}$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
✗ 3	0	0	3	4	4	7
✗ 4	0	0	3	4	5	$\boxed{7}$

Since the previous value is different, we can the current value's corresponding item

item:	1	2	3	4
		1	0	0

$$B[i,w] = \max(B[i-1,w],\ B[i-1,\ w-w[i]] + v[i])$$

$$B[4,4] = \max(B[3,4],\ B[3,4-4]+5)$$

$$= \max(4,\ B[3,0]+5)$$

$$= \max(4,\ 0+5=5)$$

4

$$B[4,5] = \max(B[3,5],\ B$$

$$= \max(B[3,5),\ B$$

$$= \max(7,\ B$$

$$= \max(7,\ 5$$

Weight : 3 2 5 4

Value/ Profits : 4 3 6 5

$n = 4$

1 kg

15 kg

3 kg

B \ W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

item:	1	2	3	4
		1	0	0

$7 - 3 = 4$

$B = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$

$= \max(B[3,4], B[3, 4-4] + 5)$

$= \max(4, B[3,0] + 5)$

$= \max(4, 0+5 = 5)$

$= \max(4, 5)$

$= 5$

$B[4,5] = \max(B[3,5], B[3, 5-4] + 5)$

$= \max(B[3,5], B[3,1] + 5)$

$= \max(7, B\ 0 + 5 = 5)$

$= \max(7, 5)$

weight : 3 2 5 4 | n = 4

Value/ Profits : 4 3 6 5

1kg

15 kg

3kg

B	i ↓ \ W →	0	1	2	3	4	5
	0	0	0	0	0	0	0
	1	0	0	0	4 ←	4	4
✓	2	0	0	3	4	4	7
	3	0	0	3	4	4	7
	4	0	0	3	4	5	$\boxed{7}$

item:	1	2	3	4
		1	0	0

7 - 3 = 4

We look for this value in the previous value's row, and set the pointer to that value

= max (... [i]] + v[i])

= max (... 5] = max(B[3, 5], B[3, 5 - 4] + 5)

= max (... max(B[3, 5), B[3, 1] + 5)

= ... max(7, B 0 + 5 = 5)

max(7, 5)

weight : 3 2 5 4

Value/ Profits : 4 3 6 5

n = 4

1kg 15 kg 3kg

item: 1 2 3 4

1 1 0 0

B i \ w

i \ w	0	1	2	3	4	5
0	0	0	0	0	0	0
✓ 1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

$7 - 3 = 4$

$= \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$

$= m \ldots + 5)$

$= m \ldots$

$= \max \ldots$

$= \max \ldots$

$B[4,5] = \max(B[3,5], B[3, 5-4] + 5)$

$= \max(B[3,5], B[3,1] + 5)$

$= \max(7, B\ 0 + 5 = 5)$

$= \max(7, 5)$

weight : 3 2 5 4 n = 4

Value / Profits : 4 3 6 5

1kg 3kg 15 kg

item: 1 2 3 4
1 1 0 0

B ($i \downarrow$, $w \rightarrow$)	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

$7 - 3 = 4$

$4 - 4 = 0$

$= \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$

$= \max(B[3,4], B[3, 4-4] + 5)$

$= \max(4, B[3,0] + 5)$

$= \max(4, 0 + 5 = 5)$

$= \max(4, 5)$

$= 5$

$B[4,5] = \max(B[3,5], B[3, 5-4] + 5)$

$= \max(B[3,5), B[3,1] + 5)$

$= \max(7, B\ 0 + 5 = 5)$

$= \max(7, 5)$

weight : 3 2 5 4

Value/ Profits : 4 3 6 5

$n = 4$

B (i ↓, w →)	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	$\boxed{7}$

item:	1	2	3	4
	1	1	0	0

$7 - 3 = 4$

$4 - 4 = 0$

$= \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$

$B[4,4] = \max(B[3,4], B[3, 4-4] + 5)$

$= \max(4, B[3,0] + 5)$

$= \max(4, 0 + 5 = 5)$

$= \max(4, 5)$

$= 5$

$B[4,5] = \max(B[3,5], B[3, 5-4] + 5)$

$= \max(B[3,5], B[3,1] + 5)$

$= \max(7, B\ 0 + 5 = 5)$

$= \max(7, 5)$

Value/ Profits: 4 3 6 5

$n = 4$

15 kg

B ($i \downarrow$, $w \rightarrow$)	0	1	2	3	4	5
0	0	0	0	0	0	0
✓ 1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
✗ 3	0	0	3	4	4	7
✗ 4	0	0	3	4	5	$\boxed{7}$

item:	1	2	3	4
	1	1	0	0

$4 + 3 = 7$

$7 - 3 = 4$

$4 - 4 = 0$

$B[i,w] = \max(B[i-1,w],\ B[$ [illegible] $,\ w - w[i]] + v[i])$

$B[4,4] = \max(B[3,4],\ B[3,$ [illegible]

$= \max(4,\ B[3,0] + 5)$

$= \max(4,\ 0+5 = 5$

$= \max(4, 5)$

[illegible] $B[4,5] = \max(B[3,5],\ B[3,\ 5-4] + 5)$

[illegible] $(B[3,5),\ B[3,1] + 5)$

[illegible] $7,\ B\ 0 + 5 = 5)$

[illegible] $7,\ 5)$